

NAME: Tanisha Gandhi

DEPARTMENT: Computer Science and Engineering (Data Science)

YEAR: Third Year

SEMESTER: VI

SAP ID: 60009210197

PROJECT NAME: CODEGENIE-AN AI BASED CODING ASSISTANT

CodeGenie is a code assistant tool designed to empower programmers, particularly beginners, by providing solutions to their coding queries. This Gradio-based application leverages CodeLLama to generate code along with explanations, simplifying the process of understanding and implementing solutions.

Key Features:

- **Code Generation:** CodeGenie addresses a core need of programmers by automatically generating code based on user queries. This functionality can significantly improve efficiency and expedite the development process.
- **Explanations for Clarity:** Accompanying the generated code, CodeGenie provides clear explanations. This enhances user understanding of the code's purpose and functionality, fostering better learning and problem-solving skills.
- **Code History Management:** CodeGenie incorporates a history function, allowing users to revisit past queries and their corresponding generated code and explanations. This facilitates review, reference, and knowledge building over time.

Benefits:

- **Improved Efficiency:** Code generation streamlines the development process, enabling programmers to focus on problem-solving and logic rather than spending time writing repetitive code.
- **Enhanced Learning:** Explanations accompanying the generated code provide valuable learning opportunities, particularly for beginners who are still grasping coding concepts.
- **Knowledge Retention:** The history feature serves as a reference point, allowing users to revisit past solutions and explanations, solidifying their understanding and knowledge base.

DEMO:

The screenshot shows a web browser window with the address bar displaying `http://127.0.0.1:7860`. The browser has two tabs: "Gradio" and "(59) WhatsApp". The main content area is a dark-themed interface with a "prompt" box on the left containing the text "write a code in C++ to perform binary search". Below the prompt box are two buttons: "Clear" and "Submit". To the right of the prompt box is an "output" box containing C++ code for a binary search function. The code is as follows:

```
```cpp
#include <iostream>
using namespace std;

// Function for binary search
int binarySearch(int arr[], int n, int key)
{
 // Initialize the left and right pointers
 int left = 0, right = n - 1;

 // While the left pointer is less than or equal to the right pointer
 while (left <= right)
 {
 // Compute the middle index of the current range
 int mid = (left + right) / 2;

 // If the key is present at the middle index, return the middle index
 if (arr[mid] == key)
 return mid;

 // If the key is less than the middle index, update the left pointer
 else if (key < arr[mid])
 }
```

Below the output box is a "Flag" button. The Windows taskbar at the bottom shows the time as 20:44 on 12-04-2024.

This screenshot shows the same web browser window after the "Submit" button was clicked. The "output" box now contains the following C++ code:

```
{
 int arr[10] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};
 int key;
 cin >> key;
 cout << binarySearch(arr, 10, key);
 return 0;
}
...

```

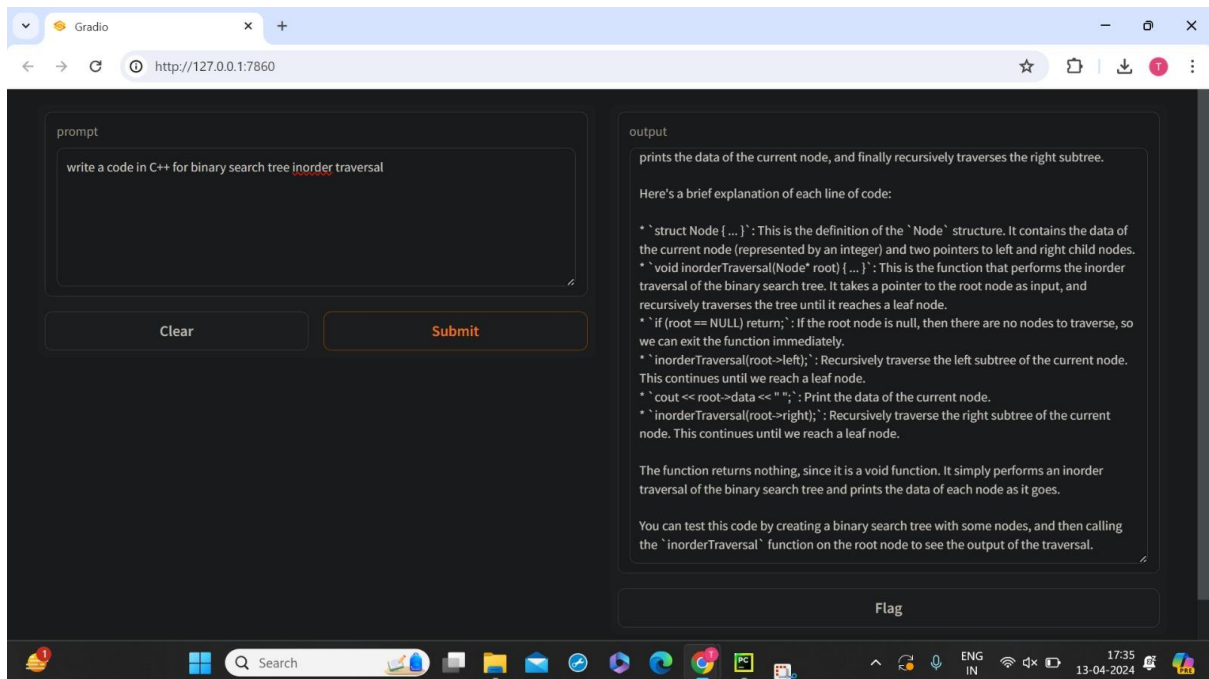
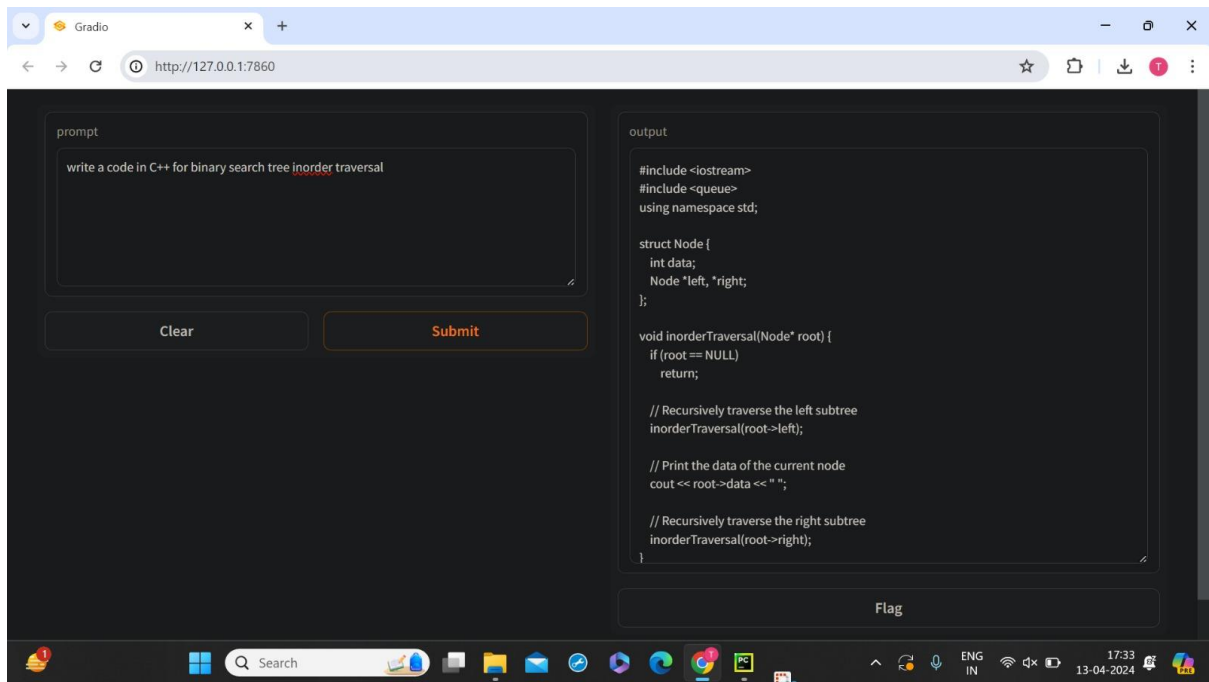
Below the code, there is a paragraph explaining the algorithm:

This code implements a binary search algorithm to search for a given 'key' in an array of integers. The function takes three arguments: the array 'arr', the length of the array 'n', and the key to be searched 'key'. It returns the index of the found key, or -1 if the key is not present in the array.

The algorithm works by iteratively dividing the search range into two halves and comparing the middle element of the range with the 'key'. If the 'key' is present at the middle index, the function returns the middle index. Otherwise, it updates the appropriate pointer to the left or right half of the range based on whether the 'key' is less than or greater than the middle element, and repeats the process until the key is found or the range reduces to a single element.

The driver code tests the binary search function by inputting a 'key' and printing the index of the found key, or -1 if the key is not present in the array.

Below the explanation is a "Flag" button. The Windows taskbar at the bottom shows the time as 20:45 on 12-04-2024.



**Github Link:** <https://github.com/tanishagandhi/Coding-Assistant>